Introduction to Wavelength Division Multiplexing

Wavelength Division Multiplexing (WDM) combines many different optical carriers (wavelengths) on a single fiber, each wavelength carrying an independent data channel.

Instead of laying many fibers, operators increase capacity by putting many wavelength channels on a single fiber.

It is conceptually the optical equivalent of frequency-division multiplexing (FDM) in radio and satellite systems: channels are separated in frequency (wavelength) to avoid interchannel interference.

Standard single-mode fibers have low-loss windows around 1310 nm and 1550 nm, spanning tens of terahertz of optical bandwidth, so many narrow channels can be packed in these windows.

WDM system design

Basic system components

Transmit side:

- Multiple transmitters (lasers), each at a different wavelength (e.g. λ1, λ2, λ3,…N).
- A **multiplexer (MUX)** that combines these wavelengths into one composite optical signal.

Fiber channel:

- Single-mode fiber carrying all wavelengths together.

Receive side:

- A **demultiplexer (DEMUX)** that separates wavelengths again.
- Receivers (photodiodes) for each wavelength.

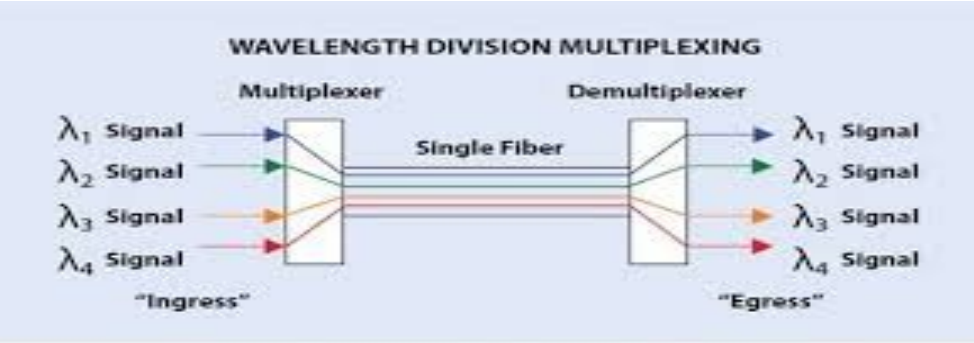


Figure 1: Basic WDM Technology Diagram

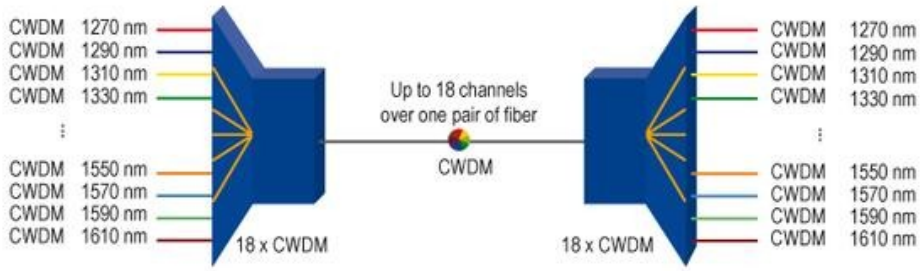
Wavelength-division Multiplexing of four channels.

WDM System Variations

WDM (or "normal" WDM): few wavelengths (e.g. 1310 nm and 1550 nm) on one fiber.

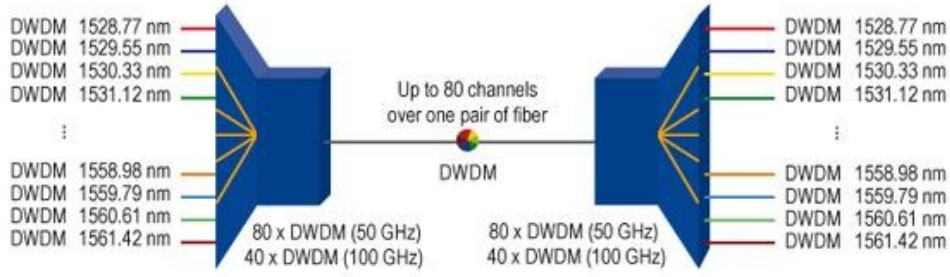
CWDM (Coarse WDM):

- Up to 18 CWDM wavelength over one pair of fiber
- CWDM channel spacing 20 nm, 1270 nm to 1610 nm
- Distances up to 120 km
- Cost-effective WDM solution
- Scalable by hybrid CWDM/DWDM



DWDM (Dense WDM):

- Up to 96 DWDM wavelength over one pair of fiber
- DWDM channel spacing 0.8 nm (100 GHz grid) or 0.4 nm (50 GHz grid)
- Distances over 1,000 km can be achieved with the use of optical amplifier
- DWDM wavelength: 1528 nm (channel 61) to 1563 nm (channel 17)



WDM Multiplexing Technology

Multiplexers and demultiplexers

A WDM multiplexer (MUX) is a passive optical device with many input ports and one (or two) fiber outputs: it combines multiple wavelength channels into one fiber without converting them back to electrical signals.

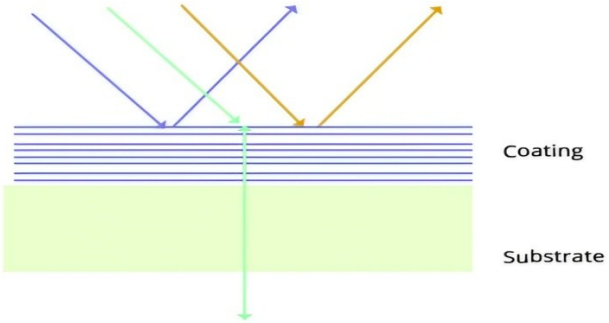
A demultiplexer (DEMUX) does the reverse: from one input fiber that carries many wavelengths, it splits out each wavelength to a separate output port, feeding individual receivers or transceivers.

Thin-film filters (TFF)

A thin-film filter is made from many very thin dielectric layers with alternating refractive indices deposited on a substrate.

Because of optical interference between reflections from each layer, some wavelengths experience constructive interference in transmission (they pass through), while others see constructive interference in reflection (they are reflected), giving a sharp passband/stopband behavior.

In WDM, vendors assemble many such filters in a tree structure: each TFF passes one wavelength (or band) and reflects the rest, so cascading them builds a MUX/DEMUX for several channels, especially in CWDM and lower-count DWDM systems.

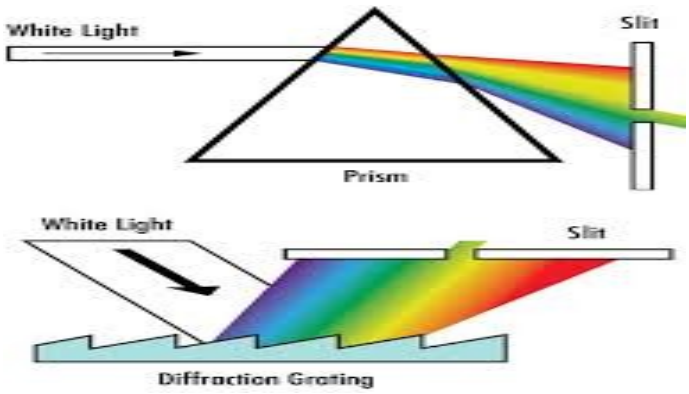


Diffraction Gratings and Prisms

A diffraction grating has many fine, periodic lines; when polychromatic light hits it, each wavelength diffracts at a different angle according to the grating equation, so the beam fans out in space into separate colors.

A prism uses material dispersion: different wavelengths have slightly different refractive indices, so they refract at different angles and emerge spatially separated.

In WDM components, lenses focus the input beam onto a grating or prism, which spreads wavelengths, and then an output fiber array or mirror arrangement picks off each wavelength to a different port, realizing a free-space MUX/DEMUX.

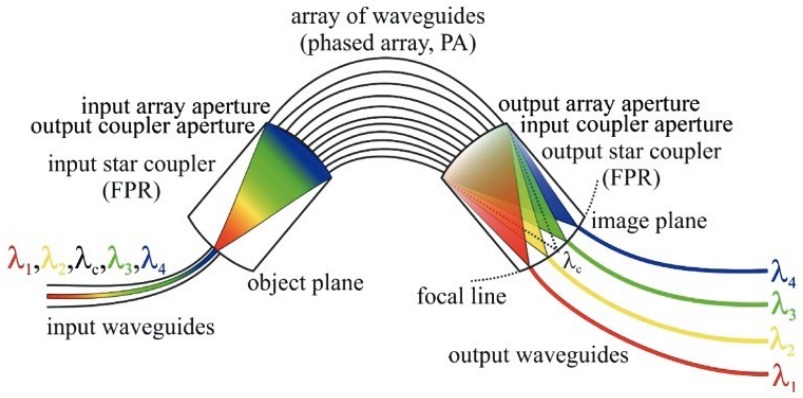


Arrayed Waveguide Gratings (AWG)

An AWG is an integrated optical circuit: it has an input waveguide, a free-propagation region, an array of many waveguides with slightly different lengths, and an output free-propagation region with multiple output waveguides.

The different path lengths in the array cause wavelength-dependent phase shifts; at the output region, interference focuses each wavelength onto a specific output port, so each port corresponds to one channel (or a small group of channels).

AWGs are widely used in dense WDM for high channel counts because they are compact, stable, and can support very small channel spacing on an integrated chip.

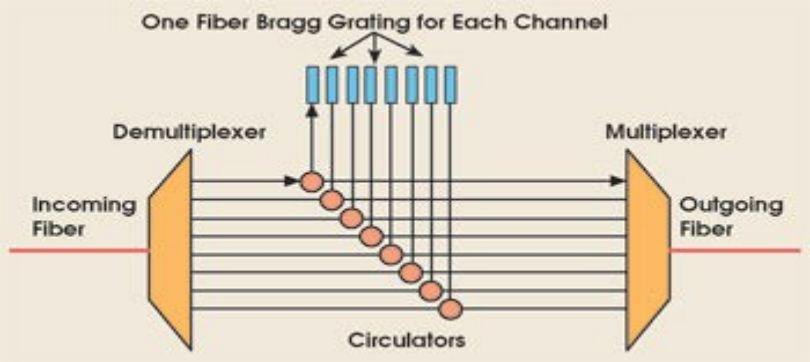


Fiber Bragg Gratings (FBG)

A fiber Bragg grating is a short segment of optical fiber where the core refractive index is periodically modulated along its length, forming a 1-D grating inside the fiber.

This grating reflects a narrow band of wavelengths (the Bragg wavelength) and transmits others; by choosing the period and length, you can design very selective reflectors or filters.

For WDM add/drop, an FBG is often used with an optical circulator: the incoming multi-wavelength signal enters port 1, the FBG reflects only the target wavelength back into the circulator, which routes that reflected channel out a different port (dropped), while all other wavelengths pass through to continue along the line; similarly, you can inject (add) a wavelength that matches the FBG's reflection band so it joins the through traffic.



Technology	Where used	Main idea
TFF	CWDM/DWDM MUX/DEMUX	Coatings reflect/pass specific λ
AWG	High-channel DWDM	Integrated waveguide interference
FBG	Add/drop filters	Grating reflects chosen λ
Gratings	Lab/special gear	Angular separation of λ

Why use WDM?

WDM is used because it lets operators multiply the capacity of a single fiber without replacing it, while avoiding the penalties of pushing each channel to very high bit rates.

1. Capacity exhaustion on existing fiber
Internet and traffic growth created demand for much more capacity on long-haul links, especially once systems moved to around 10 Gb/s per channel.

The original single-wavelength systems on those fibers ran out of capacity ("capacity exhaustion") even though the physical fiber itself could carry more if used smarter.

2. New fiber is expensive
Laying new fiber (civil works, permits, splicing, amplifiers) is slow and very costly compared with upgrading equipment at the ends of an existing fiber pair.

WDM lets you keep the same dark fiber but put multi-wavelength terminals on it, so most of the investment is in electronics/optics, not digging and cabling.

3. Limits of just increasing bit rate
Simply raising the bit rate of a single channel beyond about 2.5 Gb/s on older fibers exposes many optical impairments: chromatic dispersion, amplifier gain tilt, various nonlinearities (SPM, XPM, FWM), Raman/Brillouin scattering, polarization-mode dispersion, and reflections.

Combating all these at very high symbol rates requires expensive dispersion management, better fiber, or complex modulation/compensation, which is often harder than adding more moderate-rate wavelengths in parallel.

WDM as the practical solution
With WDM, you keep each wavelength at a manageable rate, like OC-48 (2.5 Gb/s), and run many of them together on the same fiber, so total capacity scales roughly with the number of wavelengths N.

For example, if you carry N channels at 2.5 Gb/s on one fiber, you get an aggregate of $N \times 2.5$ Gb/s without changing the fiber itself only the terminal equipment and passive MUX/DEMUX devices.

Network-level multiplexing

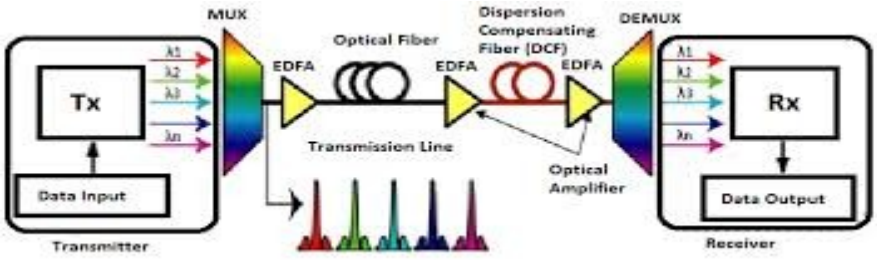
Network-level multiplexing means using WDM not just on a single fiber link, but as a network concept: wavelengths are treated like end-to-end lightpaths that connect routers, switches, or sites across a whole topology (point-to-point, ring, mesh), so capacity and routing are planned at the "wavelength" level, not only at bit level inside one fiber.

Point-to-point WDM links

This is the simplest case: one fiber pair directly connects Node A and Node B.

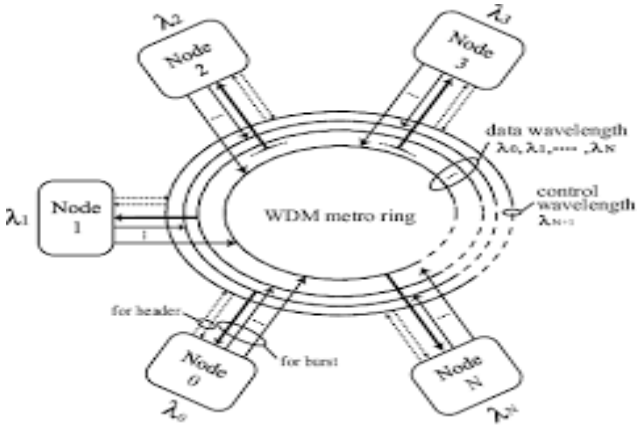
At A, multiple client signals (e.g., several 10 Gb/s or 2.5 Gb/s streams) are mapped to different wavelengths and multiplexed onto one fiber; at B, a demultiplexer splits them back to separate receivers.

The goal is just to boost capacity on that single route (long-haul backbone, DCI, etc.), with no intermediate add/drop or routing along the way.



WDM rings and mesh networks

In a WDM ring, several nodes are connected in a closed loop; many wavelengths circulate around the ring, and each node can use specific wavelengths to reach other nodes on the ring.



In a mesh WDM network, fibers interconnect nodes in a more general pattern (many possible paths); a given wavelength can be routed along a path through multiple links, giving flexible protection and routing options.

Here, wavelengths behave like circuits in the network: you can engineer which wavelength goes on which path, and use protection schemes (ring protection, mesh restoration) at the optical layer.

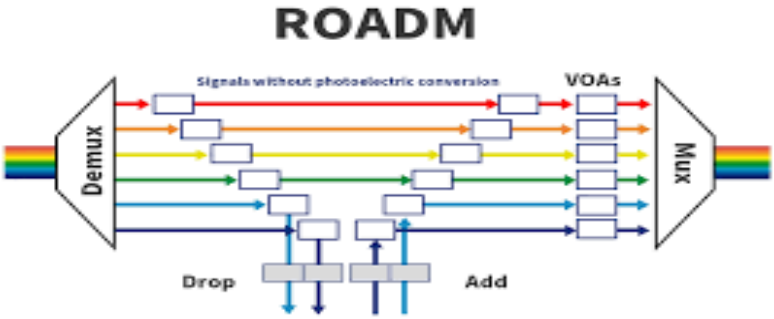
OADMs and ROADMs

A Optical Add-Drop Multiplexer (OADM) is a node that can “drop” some wavelengths from the fiber to local equipment, and “add” new wavelengths from local transmitters into the passing fiber, while letting other wavelengths bypass the node optically.

Classical OADMs are usually fixed: which wavelengths are added/dropped at a given node is determined by fixed filters/cabling; to change them you must re-patch hardware.

A Reconfigurable OADM (ROADM) does the same but under software/remote control: it uses wavelength-selective switches so the operator can dynamically choose which wavelengths to add,

drop, or pass-through, and even change paths, without converting signals to electrical form or manually rewiring.



WDM performance

WDM performance is usually discussed in terms of how much traffic you can carry and what physical factors limit quality over distance.

Capacity and scalability

Capacity formula (conceptually): total fiber capacity ≈ (number of wavelengths N) × (data rate per wavelength). If you double N or the rate per λ, the aggregate capacity roughly doubles too.

DWDM advantage: Dense WDM packs many closely spaced wavelengths (e.g. 40, 80 or more), and each channel can run at high rates such as 40 Gb/s or 100 Gb/s, giving total capacities in the multi terabit per second range on a single fiber pair.

Scalability: Operators can start with a few wavelengths and later light more λs or upgrade line rates (10G → 40G → 100G per λ) without changing the underlying fiber, so capacity grows step by step instead of in one big jump.

Key performance parameters

Attenuation and link budget

Attenuation (in dB/km) tells you how much optical power is lost per kilometre of fiber.

The link budget compares transmitter power plus amplifier gain versus all losses (fiber, connectors, MUX/DEMUX, OADMs, etc.) to ensure enough power remains at the receiver to meet the required BER.

Chromatic dispersion and dispersion slope

Chromatic dispersion makes different wavelength components of a pulse travel at slightly different speeds, so pulses broaden as distance increases, causing inter symbol interference at high bit rates.

Dispersion slope describes how dispersion changes with wavelength; in WDM, different channels see slightly different dispersion, so both the value and slope limit how far you can go at 10G/40G/100G without dispersion compensation.

OSNR (Optical Signal to Noise Ratio)

OSNR is the ratio of signal power to noise power in a given optical bandwidth at the receiver; amplifiers (EDFAs) add ASE noise that degrades OSNR each time you boost the signal.

Higher order modulation formats and higher bit rates need a higher OSNR margin; if OSNR falls too low, BER rises sharply and the channel becomes unusable.

Channel isolation / crosstalk

Channel isolation measures how well a WDM filter (MUX/DEMUX, AWG, ROADM) keeps one wavelength from leaking into a neighbouring channel.

Poor isolation leads to crosstalk, where energy from λ1 appears at λ2's receiver, increasing errors; tight filters and proper spacing keep crosstalk below specified limits.

Nonlinear effects in the fiber

At high optical power and with dense channel spacing, the fiber's nonlinear refractive index causes effects like self phase modulation (SPM), cross phase modulation (XPM), and four wave mixing (FWM).

These nonlinearities distort the signal and generate new frequency components that fall into WDM channels, effectively adding noise and crosstalk; system design must limit launch power, choose suitable fiber/dispersion, and use modulation formats that tolerate these effects.

Applications of WDM

Major telecom and Internet uses

- ✓ Long haul backbone networks: Inter city and submarine cables carrying large volumes of Internet, voice, and video traffic.
- ✓ Metro/core networks: Between central offices, PoPs, and large aggregation routers inside a region or big city.
- ✓ 5G and FTTH backhaul: High capacity links connecting mobile base stations and fiber to the home OLTs back to the core.

Data center and enterprise

- ✓ Data center interconnect (DCI): High speed links between large data centers or between buildings in a campus.
- ✓ Inside data centers (short reach WDM): Connecting switches, storage, and server clusters when parallel fibers are limited.

- ✓ Enterprise/campus networks: One pair of fibers carrying many VLANs/services (voice, data, video, backup) on different wavelengths.

Specialized

- ✓ Military and avionics: Secure, EMI immune, lightweight optical links in aircraft and defense systems.
- ✓ Cable TV / video distribution: Multiple analog/digital TV signals or IP video streams on one fiber to a hub.
- ✓ Research and testbeds: High capacity optical links in labs, universities, and experimental networks.